Finite Row-Graded Magmas over $\mathbb{F}_3$: Intrinsic Selection and Linearization

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Abstract

We study a finite family of σ-equivariant row-graded magmas over $\mathbb{F}_3$, with carrier

$$M = S \times S, \qquad S = \mathbb{F}_3,$$

left-row output, and a fixed same-row Steiner rule. The normal form is

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21}.$$

The paper has three layers. First, it maps the finite landscape: the certified associativity atlas gives

$$63 \leq \mathrm{Assoc} \leq 597,$$

with endpoint loci and exact partition function, while the global $\mathcal{H}$-frontier theorem gives

$$\min H_{tot} = -2268, \qquad \max H_{tot} = 7302, \qquad \max_{N_-=0} H_{tot} = 7020.$$

The layer above the pure frontier consists of twelve impure signed-cancellation maximizers.

Second, the central selection theorem isolates one magma by four internal finite conditions:

$$\boxed{\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J.}$$

These conditions force the right-row projection magma RP, whose multiplication is

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

Third, after selection, RP admits a linear/operator passport: rank-three left translations, a finite composition envelope, endomorphism and intertwiner structure, a binary term algebra of size

$$|T_2(\mathsf{RP})| = 630,$$

a row-contraction matrix bridge, a hidden-continuation operator identity, and companion carrier-level Lie and signed symplectic structures.

The organizing implication is

$$\boxed{\text{finite landscape}} \Longrightarrow \boxed{\text{intrinsic finite selection}} \Longrightarrow \boxed{\text{linearization of the selected magma}}.$$

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0 Introduction

0.1 Object and aim

The object of the paper is a finite landscape of row-graded multiplications on

$$M = S \times S, \qquad S = \mathbb{F}_3.$$

The first coordinate is the row and the second coordinate is the column. The family is defined by left-row output AX0, a fixed same-row off-diagonal Steiner rule AX2, and equivariance for

$$\sigma(r, c) = (r + 1, c + 1).$$

These constraints leave exactly twenty-one ternary coordinates,

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

The paper follows one spine:

$$\boxed{\text{finite row-graded landscape}} \implies \boxed{\text{intrinsic finite selection}} \implies \boxed{\text{linearization after selection}}.$$

The global atlases describe the ambient landscape. The compact selector isolates one point of this landscape. The operator, term-function, matrix-carrier, Lie, and symplectic structures then describe the selected magma.

0.2 Main theorem

Theorem 0.1 (Main theorem package)**.** Let Ω' be the row-graded landscape on $M = S \times S$, with left-row output, the fixed same-row off-diagonal Steiner rule, and σ-equivariance for $\sigma(r, c) = (r + 1, c + 1)$. Then the following hold.

(i) The landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

(ii) The finite associativity atlas satisfies

$$63 \leq \mathrm{Assoc} \leq 597,$$

with endpoint loci and exact partition function. The global $\mathcal{H}$-frontier satisfies

$$\min H_{tot} = -2268, \qquad \max H_{tot} = 7302, \qquad \max_{N_- = 0} H_{tot} = 7020.$$

The layer above the pure frontier consists of twelve impure signed-cancellation maximizers.

(iii) The four internal conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J$$

select a unique magma in Ω'. This selected magma is the right-row projection magma RP, with multiplication

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

(iv) After selection, RP has a finite linear/operator passport: every left translation has rank 3, the positive composition envelope has 51 maps, the endomorphism monoid consists of the six simultaneous symbol permutations and the three diagonal collapses, and the binary term-function algebra has

$$T_2(\mathsf{RP}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}, \qquad |T_2(\mathsf{RP})| = 630.$$

The carrier linearization

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc},$$

supports a row-contraction bridge, a hidden-continuation operator identity, carrier-level compact matrix companions, and a signed symplectic lift of the pure C/J directed-edge geometry.

(v) The proof package is finite. The normal form, information selectors, diagonal compression, and compact selection theorem are hand finite arguments. The associativity and post-selection passport data are checked against finite tables. The global $\mathcal{H}$-frontier is certified by a separate finite frontier bundle.

Proof. This theorem is the conjunction of the main results proved below. The normal form is Theorem 1.1. The associativity atlas is Theorem 5.2, with endpoint and partition data in Theorems 6.1 and 6.2. The $\mathcal{H}$-frontier is Theorem 7.2. The compact selection theorem is Theorem 12.3. The operator and term-function statements are Lemma 13.1, Theorems 14.2, 14.5, 15.2, and 17.3. The matrix, hidden-continuation operator, carrier Lie, and signed symplectic bridges are Proposition 15.4 and Theorems 16.1, 18.1, and 19.1. The finite proof interface is summarized in Section 20. □

0.3 The selected point and the second finalist

We denote the selected magma by RP, for *right-row projection.* The second local finalist is denoted by RC, for *row-complement.* The historical support alias PAB refers to RP; the mathematical text uses RP for the selected object.

The two finalists have the same global placement in the $\mathcal{H}$-frontier:

$$H_{tot} = 7020, \qquad N_- = 0.$$

They are separated by the directed-edge geometry on

$$M^\times = \{(r, c) : r \neq c\}.$$

The selected magma realizes the literal continuation/reversal pair $\{C, J\}$, while the second finalist realizes the drifted pair $\{C^{-1}, C^{-1}J\}$. The final selector is the undrifted finite C/J geometry.

0.4 Information selectors

The first two selectors are information-theoretic observables on finite random variables. The hidden-column access functional is

$$\mathcal{H}_{acc} = I(C_{out}; C_1, C_2 \mid R_1, R_2).$$

Its zero set is exactly the column-blind cross-row locus: once the ordered row pair is known, the output column is constant over the hidden input columns. This is the Landauer-facing economy selector.

The diagonal selector is the Shannon entropy of the same-point diagonal output multiset,

$$H_{diag} = -\sum_s p_s \log_2 p_s.$$

Its zero set is the constant diagonal sector. Row-anchor idempotence is the anchor condition

$$(r,r)\cdot(r,r) = (r,r) \qquad (r \in S),$$

and, after diagonal compression, it fixes the constant diagonal to $d = 000$. Equivalently, every same-point square then returns to the row anchor (r,r). The pure C/J condition completes the selection by imposing literal directed-edge head-continuation and reversal. Appendix A records the information-theoretic reading of these selectors and of the selected magma.

0.5 Landscape atlases versus selection

The global atlases and the selector have different roles. The associativity atlas and the $\mathcal{H}$-frontier atlas map the ambient finite landscape Ω'. The compact selector is a local finite theorem inside that landscape: after access-zero, diagonal entropy zero, and row-anchor idempotence, the pure C/J condition already forces RP.

A longer explanatory chain is retained because it shows the intermediate strata:

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9\times 3 \xrightarrow{a\neq b,\ d=000} 6\times 1 \xrightarrow{\min \mathrm{Assoc}_{000}} 2\times 1 \xrightarrow{\text{pure } C/J} 1.$$

This chain displays the RP/RC ambiguity before the undrifted C/J separation. The compact proof in Theorem 12.3 is the shorter logical core.

0.6 Linearization after selection

The post-selection part of the paper records the finite passport of RP. The row-owner law says that every nonempty term has output row equal to the row of its leftmost variable. The left translations have rank 3 and image a row fiber. Their positive composition envelope has 51 maps and a three-component row-target product law.

The carrier vector space has a natural matrix-unit identification

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc}.$$

Under this identification the cross-row part of RP is represented by the row-contraction bridge

$$\mu_{cross}(A,B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top, \qquad E_{ij}J_3E_{kl}^\top = E_{ik}.$$

The carrier-level Lie and signed symplectic structures are developed at this linearized level.

0.7 Proof architecture and support

The proof package uses the following finite support modes.

result family	role	support
normal form, access-zero, diagonal compression, compact RP selector	hand finite theorems	text; L2 audit tables record survivor counts
associativity range, endpoints, partition function	finite landscape atlas	`L1/` tables plus L1 verifier
global $\mathcal{H}$-frontier	finite frontier theorem	separate `mathcal_H/` bundle
operator, endomorphism, composition, T_2 structure	post-selection passport	text plus `L3/` realization and audit tables
carrier Lie and signed symplectic structures	bridge layer	`L3/` bridge audit tables

The distributed support corpus keeps the frozen finite layers and the $\mathcal{H}$-frontier certificate bundle separate:

`L1/`, `L2/`, `L3/`, `mathcal_H/`.

The reviewer-facing finite-layer audit is

```
tools/run_quick_checks.sh
```

The $\mathcal{H}$-frontier certificate bundle is kept under `mathcal_H/`; its command-level interface is recorded in `SUPPORT_INDEX.txt`.

0.8 Positioning

The selection layer uses standard finite information observables in the language of Landauer, Bennett, Shannon, and Cover–Thomas [19, 3, 4, 26, 10]. The finite magma, term-function, and clone-theoretic context is standard universal algebra [5, 8, 7, 24, 25, 27, 20]. The same-row rule is the three-point Steiner multiplication, in the classical nonassociative and quasigroup background [6, 11, 23, 9]. The certificate layer belongs to computer-assisted finite mathematics [1, 2, 13, 21, 15, 16]. The carrier-level matrix, Lie, and symplectic terminology follows standard conventions [18, 12, 17, 14, 22].

0.9 Notation

notation	meaning
S	the field $\mathbb{F}_3$
M	the carrier $S \times S$
σ	the simultaneous shift $(r, c) \mapsto (r + 1, c + 1)$
F_r	the row fiber $\{r\} \times S$
$M^\times$	the six off-diagonal points $\{(r, c) : r \neq c\}$
Ω'	the landscape satisfying AX0, AX2, and σ-equivariance
(A, B, d)	normal coordinates in $S^{18} \times S^3$
$h_{a,b}$	column-blind cross-row rule with anchors $h(0, 1) = a$, $h(0, 2) = b$
Assoc	total number of associative triples in M^3
Assoc_{000}	local associativity statistic after fixing $d = 000$
$\mathcal{H}_{acc}$	hidden-column access observable

notation	meaning
H_{diag}	entropy of the diagonal output multiset
h_q, H_{tot}, N_-	hidden-continuation local contribution, total value, and negative local count
C, J	continuation and reversal maps on $M^\times$
$T_2(\mathsf{RP})$	binary term-function algebra of RP

Part I

The finite landscape

1 Carrier, axioms, and normal form

1.1 Carrier and shift

Let

$$S = \mathbb{F}_3, \qquad M = S \times S.$$

All arithmetic in row and column coordinates is in S. For $x = (r, c) \in M$, write

$$\mathrm{row}(x) = r, \qquad \mathrm{col}(x) = c,$$

and let

$$F_r = \{r\} \times S$$

be the row fiber. We use the cyclic shift

$$\sigma(r, c) = (r + 1, c + 1).$$

Thus σ translates rows and columns simultaneously.

For distinct $i, j \in S$, write $\overline{ij}$ for the third element of $S \setminus \{i, j\}$. Equivalently, in $\mathbb{F}_3$,

$$\overline{ij} = -i - j \qquad (i \neq j).$$

1.2 Axioms

The landscape Ω' consists of binary operations $\cdot$ on M satisfying the following three conditions.

AX0, left-row output.
For all $x, y \in M$,

$$\mathrm{row}(x \cdot y) = \mathrm{row}(x).$$

Equivalently, $x \cdot y \in F_{\mathrm{row}(x)}$.

σ-equivariance.
For all $x, y \in M$,

$$\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y).$$

AX2, same-row off-diagonal Steiner rule.
For all $r, i, j \in S$ with $i \neq j$,

$$(r, r + i) \cdot (r, r + j) = (r, r + \overline{ij}).$$

Equivalently, in absolute column notation,

$$(r, c_1) \cdot (r, c_2) = (r, \overline{c_1 c_2}) \qquad (c_1 \neq c_2).$$

This is the Steiner multiplication on a three-point row fiber [23, 9].

AX0 fixes only the output row. The output column remains free in the cross-row branch and on the same-point diagonal branch.

1.3 Normal coordinates

It is convenient to use row-relative coordinates. A point of row r is written as $(r, r+i)$, where $i \in S$ is the column offset from the row.

A normal coordinate is a triple

$$(A, B, d) \in S^9 \times S^9 \times S^3,$$

where

$$A = (A_{ij})_{i,j \in S}, \qquad B = (B_{ij})_{i,j \in S}, \qquad d = (d_0, d_1, d_2).$$

The associated multiplication is defined by the four formulas

$$(r, r+i) \cdot (r+1, r+1+j) = (r, r + A_{ij}),$$
$$(r, r+i) \cdot (r+2, r+2+j) = (r, r + B_{ij}),$$
$$(r, r+i) \cdot (r, r+i) = (r, r + d_i),$$

and, for $i \neq j$,

$$(r, r+i) \cdot (r, r+j) = (r, r + \overline{ij}).$$

The first two arrays record the two ordered cross-row representatives $(0, 1)$ and $(0, 2)$. The vector d records the same-point diagonal. The fourth formula is AX2.

Conversely, any operation satisfying AX0, σ-equivariance, and AX2 has a unique coordinate triple (A, B, d): after applying a power of σ, every cross-row product reduces to one of the two row-pair representatives above, and every same-point product reduces to one of the three diagonal coordinates.

1.4 Normal form theorem

Theorem 1.1. The AX0+AX2, σ-equivariant landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Consequently

$$|\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Proof. AX0 fixes the output row of every product. AX2 fixes all same-row off-diagonal output columns. The same-point diagonal branch has three row-relative coordinates d_0, d_1, d_2. For cross-row products, σ-equivariance reduces the ordered row-pair data to the two representatives $(0, 1)$ and $(0, 2)$. Each representative has 3×3 column-offset inputs, hence contributes nine independent coordinates. Thus the cross-row branch contributes 18 coordinates and the diagonal branch contributes 3 coordinates.

The coordinate construction above gives a product satisfying AX0, σ-equivariance, and AX2 for every $(A, B, d) \in S^9 \times S^9 \times S^3$, and the preceding paragraph recovers these coordinates uniquely from any such product. Hence $\Omega' \cong S^{21}$, and its cardinality is 3^{21}. $\square$

2 Effective symmetry and orbit count

The oriented system used in the selection theorem fixes the generator $s \mapsto s+1$ through $\sigma(r,c) = (r+1, c+1)$. For landscape compression, one may also use the normalizer of this cyclic shift in $\mathrm{Sym}(S)$. Translations $s \mapsto s+\beta$ commute with the shift and act trivially on the row-relative normal coordinates. The remaining normalizer class is represented by the reflection $\nu(s) = -s$, which conjugates σ to σ^{-1}. Since σ-equivariance is equivalent to σ^{-1}-equivariance, this reflection preserves Ω' as a landscape-compression symmetry.

On normal coordinates it induces the involution

$$\iota(A, B, d) = (A', B', d')$$

with indices in S and

$$A'_{ij} = -B_{-i,-j}, \qquad B'_{ij} = -A_{-i,-j}, \qquad d'_i = -d_{-i}.$$

We denote the resulting residual effective action by

$$G_{\mathrm{eff}} = \langle \iota \rangle \cong C_2.$$

On the diagonal coordinate sector $\Delta = S^3$, the same residual reflection gives the compression

$$|\Delta / G_{\mathrm{eff}}| = \frac{3^3 + 3}{2} = 15.$$

Proposition 2.1. The effective orbit count of the full normal-coordinate landscape is

$$|\Omega' / G_{\mathrm{eff}}| = \frac{3^{21} + 3^{10}}{2}.$$

Proof. Burnside's lemma applies to the displayed involution. The identity contributes 3^{21}. For a fixed point of ι, the nine entries of A are free and then determine B by $B_{ij} = -A_{-i,-j}$. The diagonal equations are

$$d_0 = -d_0, \qquad d_1 = -d_2, \qquad d_2 = -d_1,$$

so $d_0 = 0$ and d_1 is free. Hence the fixed locus has $9 + 1 = 10$ ternary degrees of freedom and size 3^{10}. Therefore

$$|\Omega' / G_{\mathrm{eff}}| = \frac{3^{21} + 3^{10}}{2}.$$

□

The orbit count is a landscape compression. The residual reflection is not used to identify the two directed-edge absorption geometries in the selection theorem; the pure C/J maps are defined intrinsically on the directed-edge set in Section 12.

3 Hidden-column access

The first selection functional is information-theoretic. It measures whether the cross-row rule has functional dependence on hidden column coordinates after the row pair is known. Its vanishing is the Landauer-facing condition that the cross-row channel admits a realization through the row data alone.

3.1 Column-blind cross-row rules

A cross-row output rule is **column-blind** if its output column depends only on the two row coordinates:

$$g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Equivalently, after the ordered row pair is fixed, the output column is constant over the nine column inputs.

By σ-equivariance, a column-blind rule is determined by two values

$$a = h(0, 1), \qquad b = h(0, 2).$$

We call (a, b) the cross-row anchors and write $h_{a,b}$ for the corresponding rule. The remaining ordered row pairs are forced by σ-shift:

$$h(r, r + 1) = r + a, \qquad h(r, r + 2) = r + b$$

with arithmetic in $\mathbb{F}_3$. There are $3^2 = 9$ column-blind cross-row rules.

3.2 Hidden-column access as a Landauer-facing observable

Let

$$\Sigma_{cross} = \{(r_1, c_1, r_2, c_2) \in S^4 : r_1 \neq r_2\}$$

with the uniform distribution, and let

$$C_{out} = g(R_1, C_1, R_2, C_2)$$

be the output-column random variable. Define the hidden-column access observable by

$$\boxed{\mathcal{H}_{acc}(g) = I(C_{out}; C_1, C_2 \mid R_1, R_2).}$$

Logarithms are base 2, so the value is measured in bits.

This is a structural information observable. It measures how much hidden-column information is functionally retained in the cross-row output once the row pair is fixed. The Landauer reading is that hidden column information which is read and later discarded by an irreversible implementation carries thermodynamic cost [19, 3, 4]. Therefore $\mathcal{H}_{acc} = 0$ selects exactly those cross-row rules for which that cost is structurally avoidable: the output is computed through the public row interface.

The zero condition is used in this implementation-independent structural sense: the column-read burden can be avoided altogether.

3.3 Access-zero theorem

Theorem 3.1. For a cross-row rule g,

$$\mathcal{H}_{acc}(g) = 0 \quad \Longleftrightarrow \quad g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

There are exactly nine column-blind cross-row rules.

Proof. Conditional mutual information vanishes exactly when

$$C_{out} \perp (C_1, C_2) \mid (R_1, R_2).$$

Since C_{out} is a deterministic function of (R_1, C_1, R_2, C_2), conditional independence is equivalent to constancy in (C_1, C_2) for every fixed ordered row pair (R_1, R_2). That is precisely column-blindness.

The two representative cross-row row pairs $(0, 1)$ and $(0, 2)$ each receive one output-column value in S, giving $3^2 = 9$ σ-equivariant column-blind rules. □

This theorem is a landscape result and also the first reduction in the selection chain.

4 Associativity master formula

4.1 The associativity statistic

For $x \in \Omega'$, define

$$\mathrm{Assoc}(x) = \#\{(u, v, w) \in M^3 : (u \cdot v) \cdot w = u \cdot (v \cdot w)\}.$$

Since $|M| = 9$, this statistic lies between 0 and 729.

Theorem 4.1. In normal coordinates,

$$\mathrm{Assoc} = 3\, rawAssoc,$$

where $rawAssoc$ is a normalized sum over $3^5 = 243$ triples and decomposes as

$$rawAssoc = rawRRR + rawRRS + rawRSR + rawSRR + rawDIST.$$

The five blocks correspond to the row patterns

$$RRR, \qquad RRS, \qquad RSR, \qquad SRR, \qquad DIST.$$

Proof. By σ-equivariance the first row coordinate of a triple may be normalized to 0. The two remaining row variables and the three column variables give 3^5 normalized cases; the factor 3 restores the three translates. The displayed blocks are the five equality patterns of the three rows. □

4.2 The same-row block

The all-same-row block depends only on the diagonal vector d. Let

$$E(d) = \#\{(i, j) : i \neq j,\ d_i = d_j\}, \qquad N(d) = \#\{(i, j) : i \neq j,\ d_i = j,\ d_j = j\}.$$

Lemma 4.2. The same-row contribution is

$$rawRRR(d) = 9 + E(d) + 2N(d).$$

The four diagonal classes give raw same-row values

$$9, \qquad 11, \qquad 13, \qquad 19.$$

In particular,

$$rawRRR(000) = 19.$$

Proof. In one row, unequal column pairs use the Steiner rule and equal pairs use d. The 27 triples split into a baseline family plus the two diagonal-sensitive families counted by $E(d)$ and $N(d)$. This gives the formula and the four class values. □

4.3 Column-blind profile at $d = 000$

Let $h_{a,b}$ be the column-blind rule

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b.$$

At $d = 000$, same-point products satisfy $(r, c)^2 = (r, r)$.

Proposition 4.3. On the column-blind, $d = 000$ stratum,

$$\begin{aligned}(rawRRR, rawRRS, rawRSR, rawSRR, rawDIST)(a, b, 000)\\ = \big(19,\ 9[a=0] + 9[b=0],\ 9[a=0] + 9[b=0],\\ 54,\ 54[a=b]\big).\end{aligned}$$

Consequently

$$\boxed{\mathrm{Assoc}_{000}(h_{a,b}) = 3\,(73 + 18[a=0] + 18[b=0] + 54[a=b])\,.}$$

Proof. The RRR value is Lemma 4.2. In the RRS and RSR patterns, the same-row step must return the cross-row anchor; at $d = 000$ this contributes 9 exactly when the relevant anchor is 0. In SRR, the final cross-row output ignores same-row column processing, giving all 54 normalized triples. In $DIST$, the two association orders compare the two cross-row anchors, so all 54 triples survive exactly when $a = b$. □

The nine column-blind rules split as follows:

class	representatives	profile	Assoc_{000}
anchored trivial	$(0, 0)$	$(19, 18, 18, 54, 54)$	489
off-anchor trivial	$(1, 1), (2, 2)$	$(19, 0, 0, 54, 54)$	381
partly anchored nondegenerate	$(0, 1), (0, 2), (1, 0), (2, 0)$	$(19, 9, 9, 54, 0)$	273
fully off-anchor nondegenerate	$(1, 2), (2, 1)$	$(19, 0, 0, 54, 0)$	219

The fourth class consists of RP and RC.

5 Global associativity range

5.1 Exact range

Lemma 5.1 (Associativity atlas certificate protocol)**.** The global associativity atlas is used through five finite obligations.

1. **A1, block reduction.** The associativity statistic is reduced by σ-normalization to the five row-pattern blocks

$$rawAssoc = rawRRR + rawRRS + rawRSR + rawSRR + rawDIST$$

of Theorem 4.1.

2. **A2, diagonal decomposition.** The same-row block is evaluated from the diagonal vector by Lemma 4.2, giving the four possible values

$$9, \quad 11, \quad 13, \quad 19.$$

3. **A3, fixed-diagonal cross-range certificates.** For each diagonal class, the certificate verifies the exact range of

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

4. **A4, endpoint witnesses and endpoint counts.** The endpoint certificates verify the normal-coordinate witnesses, endpoint counts, and orbit representatives for Assoc = 63 and Assoc = 597.
5. **A5, distribution consistency.** The partition-function certificate gives the full coefficient table for

$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$

and checks total mass, range, moments, and mode.

Any certificate satisfying A1–A5 proves the global associativity range, endpoint loci, and partition-function claims of Sections 5–6. This is a certified atlas statement for the ambient landscape; the compact RP uniqueness proof in Theorem 12.3 is the separate hand spine in Part II. □

Theorem 5.2 (Global associativity atlas)**.** On Ω',

$$21 \leq rawAssoc \leq 199.$$

Equivalently,

$$63 \leq \mathrm{Assoc} \leq 597.$$

The finite certificate is organized by diagonal class:

diagonal class	rawRRR	certified rawCross range	rawAssoc consequence
permutation	9	12..180	21..189
two-valued, no fixed point	11	10..162	21..173
two-valued, with fixed point	13	12..174	25..187
constant	19	12..180	31..199

Here

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

The endpoint values are therefore $rawAssoc = 21$ and $rawAssoc = 199$, or Assoc = 63 and Assoc = 597.

Proof. The hand-readable reduction is Theorem 4.1 plus Lemma 4.2. After that reduction, the global claim is exactly the A3 cross-range obligation of Lemma 5.1. Adding the certified cross-row ranges to the four possible same-row values gives the displayed global range.

The endpoint witnesses, endpoint counts, orbit representatives, and partition-function consistency are the A4–A5 obligations of Lemma 5.1. They are recorded in the `L1/tables/` atlas files listed in `SUPPORT_INDEX.txt` and checked by `L1/scripts/verify_layer1_v3_final.py`. □

5.2 Global and local associativity roles

RP has

$$\mathrm{Assoc}(\mathsf{RP}) = 219,$$

while the landscape minimum is 63. The global statistic maps the ambient landscape. The selection argument later uses the conditional local statistic Assoc_{000}, after information compression and anchoring, to expose the final two-candidate ambiguity.

6 Endpoint loci and partition function

6.1 Endpoint loci

Theorem 6.1. The associativity endpoint loci have sizes

$$|\mathcal{M}_{63}| = 24, \qquad |\mathcal{M}_{63}/\operatorname{Sym}(S)| = 12,$$

and

$$|\mathcal{M}_{597}| = 6, \qquad |\mathcal{M}_{597}/\operatorname{Sym}(S)| = 3.$$

The minimum endpoint block profiles are

$$(11, 2, 2, 2, 4) \quad \text{and} \quad (9, 2, 2, 4, 4),$$

while the maximum endpoint block profile is

$$(19, 36, 36, 54, 54).$$

Proof. This is the A4 endpoint part of Lemma 5.1. The endpoint certificates record the exact solution counts by diagonal sector, orbit representatives, and explicit witnesses in normal coordinates. The minimum occurs in the permutation and two-valued/no-fixed diagonal classes. The maximum occurs in the constant diagonal class. The block profiles are obtained by evaluating the five normalized associativity blocks on the endpoint witnesses. □

6.2 Exact associativity partition function

Theorem 6.2. The exact polynomial

$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$

has 167 nonzero coefficients and total mass

$$Z(1) = 3^{21}.$$

Its first two moments are

$$\mathbb{E}[\mathrm{Assoc}] = 245, \qquad \mathrm{Var}(\mathrm{Assoc}) = \frac{27820}{27}.$$

The mode occurs at

$$\mathrm{Assoc} = 237$$

with coefficient

$$426, 317, 976.$$

Proof. This is the A5 partition-function part of Lemma 5.1. The coefficient certificate in `L1/tables/` is the exact finite distribution of Assoc on Ω'. The total mass, range, mean, variance, and mode are computed directly from that table and checked by `L1/scripts/verify_layer1_v3_final.py`. □

7 Hidden continuation and the $\mathcal{H}$-frontier

7.1 Local distance reduction

The hidden-continuation observable is a finite path-information defect. The continuation side keeps the two-step route data, while the left-translation side compares only the collapsed endpoint operators. Thus $h_q/2$ is the signed Hamming contrast between a path-resolved comparison and an endpoint-resolved comparison. Negative local values are precisely the local cancellation failures counted by N_-. The normalization $H_{tot} = 6rawH$ is the σ-orbit normalization used by the $\mathcal{H}$-frontier certificate.

Its local index is

$$q = (b, t, a, e, f) \in S^5.$$

For a completed normal-form point $x \in \Omega'$, let M_s denote the local table read: M_1 and M_2 are the two cross-row tables, and M_0 is the same-row diagonal/complement rule. All row and column arithmetic below is in $S = \mathbb{F}_3$. Define

$$xy = M_b(a, e), \qquad yz = M_{t-b}(e - b, f - b), \qquad z_R = b + yz,$$

and

$$ep_L = M_t(xy, f), \qquad ep_R = M_b(a, z_R).$$

For $c \in S$, define the collapsed left-translation signature

$$L_c(i, j) = M_i(c, j) \qquad ((i, j) \in S^2).$$

For $s, \alpha, z \in S$, define the stepwise continuation signature

$$C_{s,\alpha,z}(i, j) = M_s(\alpha,\ s + M_{i-s}(z - s, j - s)) \qquad ((i, j) \in S^2).$$

The associated Hamming distances are

$$D_L(c, c') = \#\{(i, j) \in S^2 : L_c(i, j) \neq L_{c'}(i, j)\},$$

and

$$D_C(s, \alpha, z; s', \alpha', z') = \#\{(i, j) \in S^2 : C_{s,\alpha,z}(i, j) \neq C_{s',\alpha',z'}(i, j)\}.$$

Direct local definition. For each of the nine test pairs $(i, j) \in S^2$, set

$$\delta^C_{ij}(q; x) = \mathbf{1}\{C_{t,xy,f}(i, j) \neq C_{b,a,z_R}(i, j)\}\,,$$

and

$$\delta^L_{ij}(q; x) = \mathbf{1}\{L_{ep_L}(i, j) \neq L_{ep_R}(i, j)\}\,.$$

The nine direct local summands are

$$\Delta_{ij}(q; x) = 2\big(\delta^C_{ij}(q; x) - \delta^L_{ij}(q; x)\big),$$

displayed as

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	Δ_{00}	Δ_{01}	Δ_{02}
$i = 1$	Δ_{10}	Δ_{11}	Δ_{12}
$i = 2$	Δ_{20}	Δ_{21}	Δ_{22}

and the direct hidden-continuation contribution is

$$h_q(x) = \sum_{(i,j)\in S^2} \Delta_{ij}(q; x).$$

Theorem 7.1 (Local distance reduction)**.** For every $x \in \Omega'$ and $q \in S^5$,

$$\boxed{h_q(x) = 2\left(D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R)\right).}$$

Consequently the local direct definition and the Hamming-distance formula define the same global observables $rawH$ and H_{tot}.

Proof. By definition, $D_C(t, xy, f; b, a, z_R)$ is the sum of the nine indicators $\delta^C_{ij}(q; x)$, and $D_L(ep_L, ep_R)$ is the sum of the nine indicators $\delta^L_{ij}(q; x)$. Subtracting these two Hamming distances and multiplying by 2 gives exactly the displayed sum of the nine direct local summands Δ_{ij}. The symbolic reduction is recorded by the $\mathcal{H}$-frontier certificate handles `HF-RED` and `HF-CERT-IF`. □

Then

$$rawH(x) = \sum_{q \in S^5} \frac{h_q(x)}{2}, \qquad H_{tot}(x) = 6\, rawH(x) = 3 \sum_{q \in S^5} h_q(x),$$

and

$$\boxed{N_-(x) = 3\,\#\{q \in S^5 : h_q(x) < 0\}.}$$

Thus purity is the finite system

$$N_-(x) = 0 \quad \Longleftrightarrow \quad h_q(x) \geq 0 \quad (q \in S^5).$$

7.2 Global frontier theorem

The global $\mathcal{H}$-frontier theorem is supported by four finite certificate obligations.

handle	obligation	result
H1	unrestricted full-landscape range	$rawH \in [-378, 1217]$, endpoint counts $8, 12$
H2	pure-frontier exclusion	$(h_q \geq 0\ \forall q) \wedge rawH \geq 1171$ is UNSAT
H3	pure-frontier witnesses	RP and RC have $rawH = 1170$, $N_- = 0$
H4	signed-cancellation spike	exactly twelve points above the pure frontier, all with $(rawH, N_-) = (1217, 3)$

Theorem 7.2 (Global frontier of $\mathcal{H}$)**.** On $\Omega' = S^{21}$:

1. The unrestricted range is

$$\boxed{\min H_{tot} = -2268, \qquad \max H_{tot} = 7302.}$$

   Equivalently, $rawH \in [-378, 1217]$. The endpoint multiplicities are 8 and 12.

2. The pure frontier is

$$\boxed{\max\{H_{tot} : N_- = 0\} = 7020.}$$

3. The region above the pure frontier is exactly the unrestricted maximum layer:

$$\boxed{\{H_{tot} > 7020\} = \{H_{tot} = 7302\}.}$$

This layer has 12 points, all with

$$rawH = 1217, \qquad H_{tot} = 7302, \qquad N_- = 3,$$

$$H_{pos} = 7308, \qquad H_{\text{neg_abs}} = 6, \qquad h_{\text{loc,min}} = -2, \qquad h_{\text{loc,max}} = 18.$$

Proof. The analytic reduction is Theorem 7.1. H1 covers all 3^{21} normal-form words and gives the unrestricted range. For the pure frontier, $N_- = 0$ is the system $h_q \geq 0$ for all $q \in S^5$, and $H_{tot} > 7020$ is equivalent to $rawH \geq 1171$. H2 proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

UNSAT on all 18540 depth-9 live nodes:

$$18540/18540 \text{ UNSAT}, \qquad SAT = 0, \qquad UNKNOWN = 0.$$

H3 supplies pure witnesses at $rawH = 1170$, and H4 classifies the full layer above the pure threshold as the twelve impure maximizers. □

7.3 RP and RC on the pure frontier

Corollary 7.3. RP and RC satisfy

$$H_{tot} = 7020, \qquad rawH = 1170, \qquad N_- = 0.$$

Thus both lie on the global pure $\mathcal{H}$-frontier.

Proof. Direct evaluation of

$$\mathsf{RP} = 11111111122222222000, \qquad \mathsf{RC} = 22222222211111111000$$

gives the displayed values, and Theorem 7.2 proves that no pure point has larger H_{tot}. □

Part I has now supplied the ambient finite map: normal form, global associativity, and the $\mathcal{H}$-frontier. Part II turns from atlas data to the intrinsic selector. Its proof uses the access-zero and diagonal-compression reductions, then the directed-edge absorption geometry; the frontier theorem supplies the global placement of the two selection finalists.

Part II

Intrinsic finite selection

8 Selection core

The selection theorem is the finite hand spine of the paper. It uses four internal conditions:

$$\boxed{\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J.}$$

Their order follows the row/column geometry of Ω'. The first condition makes the cross-row branch column-blind, so it has anchors $(a, b) \in S^2$. The second condition compresses the same-point diagonal to a constant triple, and row-anchor idempotence fixes this triple to $d = 000$. At that point the absorption geometry on the six directed edges $M^\times$ reads the remaining cross-row anchors directly; the literal continuation/reversal criterion forces $(a, b) = (1, 2)$. Thus the compact core selects the right-row projection magma RP.

In normal coordinates, the selected point and the second diagnostic finalist are

$$\mathsf{RP} = (a, b, d) = (1, 2, 000), \qquad \mathsf{RC} = (a, b, d) = (2, 1, 000).$$

The distinction between them is geometric: RP realizes the undrifted pair $\{C, J\}$ on directed edges, whereas RC realizes $\{C^{-1}, C^{-1}J\}$.

The longer explanatory chain is retained as a diagnostic path through the same finite landscape:

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9 \times 3 \xrightarrow{a\neq b,\ d=000} 6 \times 1 \xrightarrow{\min \mathrm{Assoc}_{000}} 2 \times 1 \xrightarrow{\text{pure } C/J} 1.$$

It records how local path-dependence exposes the RP/RC ambiguity before the undrifted C/J criterion separates the two finalists. The logical core remains Theorem 12.3.

9 Diagonal compression

The access-zero theorem has already compressed the cross-row sector to the nine column-blind rules $h_{a,b}$. The remaining information-compression step acts on the same-point diagonal data.

For a diagonal triple

$$d = (d_0, d_1, d_2) \in S^3,$$

let

$$p_s = \frac{1}{3}\#\{i : d_i = s\}.$$

Define the diagonal Shannon observable by

$$\boxed{H_{diag}(d) = -\sum_{s\in S} p_s \log_2 p_s.}$$

The possible levels are:

diagonal type	multiplicities	H_{diag}
constant	$(3, 0, 0)$	0
two-valued	$(2, 1, 0)$	$\log_2 3 - \frac{2}{3}$
permutation	$(1, 1, 1)$	$\log_2 3$

Theorem 9.1. For diagonal triples,

$$H_{diag}(d) = 0 \quad \Longleftrightarrow \quad d_0 = d_1 = d_2.$$

Thus diagonal compression reduces the diagonal sector from 27 choices to

$$000, \qquad 111, \qquad 222.$$

Proof. Entropy of a finite distribution vanishes exactly when the distribution is concentrated on one symbol. For the empirical distribution of (d_0, d_1, d_2), this is precisely constancy of the diagonal triple. □

Together with the access-zero theorem of Section 3, this reduces the selection domain to 9×3.

10 Diagnostic anchors and local path-dependence

This section records the explanatory branch of the selection story. The condition $a \neq b$ and the local statistic Assoc_{000} are useful diagnostics for the finite landscape: they expose the two nearby finalists before the directed-edge criterion makes the final choice.

10.1 Nondegenerate diagnostic anchors

Proposition 10.1. Among column-blind cross-row rules $h_{a,b}$, the condition

$$a \neq b$$

leaves exactly six rules.

Proof. There are nine pairs $(a, b) \in S^2$. The three diagonal pairs (a, a) identify the two cross-row representatives under a single anchor. The remaining $3^2 - 3 = 6$ pairs are the nondegenerate rules. □

Proposition 10.2. After diagonal compression, row-anchor idempotence selects the unique constant diagonal

$$d = 000.$$

Proof. The compressed diagonal sector is $\{000, 111, 222\}$. Row-anchor idempotence is the condition

$$(r, r) \cdot (r, r) = (r, r) \qquad (r \in S).$$

On the compressed diagonal sector $d = sss$, this condition gives $s = 0$. Thus the unique compressed diagonal compatible with row-anchor idempotence is $d = 000$. □

10.2 Local associativity at $d = 000$

Proposition 10.3. With $d = 000$ fixed, the six nontrivial column-blind rules have local associativity values

$$\mathrm{Assoc}_{000} = 219$$

for exactly two rules, and

$$\mathrm{Assoc}_{000} = 273$$

for the remaining four. The two local minima are RP and RC.

The local-minimum profile is

$$(19, 0, 0, 54, 0),$$

whereas the four discarded rules have profile

$$(19, 9, 9, 54, 0).$$

The difference comes from the additional RRS and RSR coincidences in the discarded rules.

Proof. Proposition 4.3 gives

$$\mathrm{Assoc}_{000}(h_{a,b}) = 3\left(73 + 18[a = 0] + 18[b = 0] + 54[a = b]\right).$$

On the nondegenerate stratum $a \neq b$, this value is minimized exactly when neither anchor is 0, i.e. at $(a, b) = (1, 2)$ and $(2, 1)$. These are RP and RC, and both have value 219. The four remaining nondegenerate pairs contain one zero anchor and have value 273. The corresponding `L2/tables/` path-dependence table and `L2/scripts/verify_layer2_selection_final.py` record the same six-row audit. □

This step belongs to the diagnostic chain. It shows why the two rules RP and RC are the natural local finalists after information compression and diagonal anchoring.

11 Directed-edge geometry

Let

$$M^\times = \{(r,c) \in M : r \neq c\}$$

be the six-point directed-edge set. Each $(r,c) \in M^\times$ is interpreted as the directed edge from r to c in the triangle on vertex set S.

Define

$$C(r,c) = (c, \overline{rc}), \qquad J(r,c) = (c,r).$$

The map C advances a directed edge by making the old head the new tail and passing through the third vertex, while J reverses the directed edge.

Theorem 11.1. On $M^\times$,

$$C^3 = 1, \qquad J^2 = 1, \qquad JCJ = C^{-1}.$$

Thus C and J generate the dihedral geometry of the directed triangle.

Proof. Applying C three times traverses

$$(r,c) \mapsto (c, \overline{rc}) \mapsto (\overline{rc}, r) \mapsto (r,c).$$

Applying J twice reverses the directed edge twice. Finally,

$$JCJ(r,c) = JC(c,r) = J(r, \overline{rc}) = (\overline{rc}, r) = C^{-1}(r,c).$$

The `L3/` directed-edge audit tables record these finite relation checks. □

12 Pure C/J and the selection theorem

12.1 Intrinsicity of the pure C/J criterion

The six-point set $M^\times$ is canonically the directed-edge set of the three-vertex row alphabet. For an edge $x = (r,c)$, call r its tail, c its head, and $t = \overline{rc}$ the third vertex. Then

$$C(r,c) = (c,t), \qquad J(r,c) = (c,r).$$

Thus

$$\{C(x), J(x)\} = \{y \in M^\times : \mathrm{tail}(y) = \mathrm{head}(x)\}.$$

The pure C/J condition is therefore the head-row absorption rule: for every directed edge x, the absorbed neighbours are exactly the two directed edges whose tail is the head of x.

This is not a choice of an external cyclic orientation. For every relabelling $\rho \in \mathrm{Sym}(S)$, acting on directed edges by $\rho_\times(r,c) = (\rho(r), \rho(c))$, one has

$$\rho_\times C = C\rho_\times, \qquad \rho_\times J = J\rho_\times.$$

Indeed, ρ sends the third vertex of $\{r,c\}$ to the third vertex of $\{\rho(r), \rho(c)\}$. Hence the undrifted pair $\{C, J\}$ is an intrinsic directed-edge structure. The drifted pair $\{C^{-1}, C^{-1}J\}$ is genuinely different: its two outputs have tail $\overline{rc}$, the third vertex, rather than tail c, the head of (r,c).

12.2 Absorption and absorption transitions

For $x \in M^\times$, define the absorption set

$$\mathrm{Abs}(x) = \{y \in M^\times \setminus \{x\} : x \cdot y = x\}.$$

For a column-blind rule $h_{a,b}$ at $d = 000$, same-row right arguments never absorb an off-diagonal x. Thus absorption is governed by the cross-row equation

$$h(r_x, r_y) = c_x.$$

A candidate has a two-transition absorption geometry if every $x \in M^\times$ has exactly two absorbed neighbours. In that case the two absorbed-neighbour maps define AbsTrans. The pure C/J criterion requires

$$\mathrm{AbsTrans} = \{C, J\}$$

as literal maps on $M^\times$, i.e. absorption follows the head-row continuation/reversal pair.

12.3 The two diagnostic finalists

After local path-dependence, the two diagnostic finalists are RP and RC.

Lemma 12.1. For RP,

$$\mathrm{AbsTrans}_{\mathsf{RP}} = \{C, J\}.$$

For RC,

$$\mathrm{AbsTrans}_{\mathsf{RC}} = \{C^{-1}, C^{-1}J\}.$$

Proof. Let $x = (r, c) \in M^\times$.

For RP, $h(r_x, r_y) = r_y$. The absorption equation is

$$r_y = c.$$

Hence the two absorbed neighbours of (r, c) are the two off-diagonal points in row c:

$$(c, r), \qquad (c, \overline{rc}).$$

These are precisely

$$J(r, c) = (c, r), \qquad C(r, c) = (c, \overline{rc}).$$

So $\mathrm{AbsTrans}_{\mathsf{RP}} = \{C, J\}$.

For RC, $h(r_x, r_y) = \overline{r_x r_y}$. The absorption equation is

$$\overline{r r_y} = c,$$

so

$$r_y = \overline{rc}.$$

The two absorbed neighbours are therefore

$$(\overline{rc}, r), \qquad (\overline{rc}, c),$$

which are

$$C^{-1}(r, c) = (\overline{rc}, r), \qquad C^{-1}J(r, c) = (\overline{rc}, c).$$

Thus $\mathrm{AbsTrans}_{\mathsf{RC}} = \{C^{-1}, C^{-1}J\}$. □

12.4 Diagnostic separation of the two local finalists

Theorem 12.2. Among the two diagnostic local finalists, the pure C/J directed-edge criterion selects RP.

Proof. By Lemma 12.1, RP realizes the undrifted pair $\{C, J\}$. RC realizes $\{C^{-1}, C^{-1}J\}$, whose continuation and reversal are drifted by C^{-1}. The pure criterion selects the literal undrifted pair. The `L2/` pure-C/J survivor and transition-factorization audit tables record the same comparison. □

12.5 Compact uniqueness core

Theorem 12.3. The four conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J$$

select RP uniquely inside Ω'.

Proof. The first condition gives column-blindness by Theorem 3.1. Hence the cross-row rule has the form

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b$$

for a unique pair $(a, b) \in S^2$. The second condition gives a constant diagonal by Theorem 9.1, and row-anchor idempotence fixes this constant to $d = 000$ by Proposition 10.2.

The remaining condition is pure C/J. Let $x = (r, c) \in M^\times$ and write $\delta = c - r \in \{1, 2\}$. At $d = 000$, absorption of an off-diagonal point is governed by the cross-row equation

$$x \cdot y = x \quad \Longleftrightarrow \quad h(r, r_y) = c.$$

For the two possible cross rows this becomes

$$r_y = r + 1 \Rightarrow a = \delta, \qquad r_y = r + 2 \Rightarrow b = \delta.$$

The pure C/J criterion says that the two absorbed neighbours of $x = (r, c)$ are exactly

$$J(r, c) = (c, r), \qquad C(r, c) = (c, \overline{rc}).$$

Both lie in row c. Therefore, when $\delta = 1$, absorption must occur in row $r + 1 = c$, forcing $a = 1$. When $\delta = 2$, absorption must occur in row $r + 2 = c$, forcing $b = 2$. Thus

$$(a, b) = (1, 2).$$

This is precisely the RP cross-row rule, and $d = 000$ has already been fixed. Hence the four internal conditions select RP uniquely. □

The audit table `l2_minimal_selector_sets.csv` records the same compact core as a finite audit of the survivor condition.

12.6 Diagnostic explanatory selection chain

Proposition 12.4. The finite diagnostic chain

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9 \times 3 \xrightarrow{a\neq b,\ d=000} 6 \times 1 \xrightarrow{\min \mathrm{Assoc}_{000}} 2 \times 1 \xrightarrow{\text{pure } C/J} 1$$

has terminal survivor RP.

Proof. The first arrow is Theorems 3.1 and 9.1. The second arrow is Propositions 10.1 and 10.2. The third arrow is Proposition 10.3. The final arrow is Theorem 12.2. The audit table `l2_selection_chain.csv` records the survivor count at each reduction, and the compact L2 verifier checks the displayed chain. □

This diagnostic chain is logically subsumed by Theorem 12.3. After column-blindness, diagonal compression, and row-anchor idempotence, the pure C/J condition itself forces $(a, b) = (1, 2)$. The $\min \mathrm{Assoc}_{000}$ arrow records the local path-dependence phenomenon that exposes the RP/RC ambiguity before the undrifted directed-edge criterion separates them.

Part III

Linear/operator passport after selection

This part records the finite and carrier-level structures exposed after the selection theorem. The role of the notation is now fixed: RP is the selected right-row projection magma, and RC is the second local finalist used for comparison. The post-selection layer is a passport for the fixed selected multiplication: it records the operator, term-function, matrix-carrier, and directed-edge structures forced by RP.

The passport is organized as follows.

layer	main invariant	support
row-owner law	every nonempty term has the row of its leftmost variable	structural induction
left translations	rank three, image a row fiber, nine-dimensional span	L3 operator audit
composition envelope	$\lvert \mathrm{Mon}^+\langle L_x\rangle \rvert = 51$, split into three row-target components	L3 composition tables
endomorphisms	six simultaneous permutations and three diagonal collapses	L3 intertwiner audit
row contraction	$E_{ij}J_3E_{kl}^\top = E_{ik}$, matching the cross-row RP rule	L3 matrix bridge
binary terms	$\lvert T_2(\mathsf{RP})\rvert = 630$, with owner/same-row/cross-row factorization	L3 term tables
carrier bridges	matrix/Lie package and signed projective symplectic lift of pure C/J	L3 bridge audits

13 The selected operation and row-owner law

13.1 Selected multiplication and linearization conventions

The selected multiplication is

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

The first case is the right-row projection branch; the second is the fixed same-row Steiner branch; the third is diagonal anchoring.

The finite magma statements are set-theoretic. When the carrier is linearized, let k be a ground field and let $k[M]$ be the vector space with basis e_x, $x \in M$. Rank and image statements for individual finite maps are valid over any ground field. Linear span statements are over the chosen k. Spectral, Jordan, and fiber-algebra statements involving $t \pm 1$ or idempotents with coefficients $1/2$ are stated under

$$\operatorname{char} k \neq 2.$$

The carrier-level matrix/Lie and signed symplectic bridge statements are over $\mathbb{R}$ unless stated otherwise.

The carrier vector space has the matrix-unit identification

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc}.$$

This identification is used for carrier-level linear structures. The magma multiplication remains the finite operation displayed above.

13.2 Row-owner principle

For a nonempty term $t(x_1, \ldots, x_n)$, define its **owner** $\mathrm{ow}(t)$ to be the leftmost variable appearing in the term tree.

Lemma 13.1 (Row-owner lemma)**.** For every RP term $t(x_1, \ldots, x_n)$,

$$row\big(t(x_1, \ldots, x_n)\big) = row\big(\mathrm{ow}(t)\big).$$

Proof. The assertion is immediate for a variable. If $t = u \cdot v$, then the left-row output axiom gives $row(t) = row(u \cdot v) = row(u)$. The induction hypothesis identifies this row with the row of the leftmost variable of u, which is also the leftmost variable of $u \cdot v$. □

The row-owner lemma is the common source of the row-target product law for left-translation compositions and the owner coordinate in the binary term algebra.

14 Operator passport

For $x \in M$, write $L_x(y) = x \cdot y$.

14.1 Left translations

Proposition 14.1. For $x = (r, c)$,

$$L_{(r,c)}(s,d) = \begin{cases} (r,s), & s \neq r, \\ (r,r), & s = r,\ d = c, \\ (r,\overline{cd}), & s = r,\ d \neq c. \end{cases}$$

Consequently $L_{(r,c)}(M) = F_r$.

Proof. The three cases are exactly the cross-row projection, diagonal anchoring, and same-row Steiner branches of RP. They produce all three points of F_r. □

Theorem 14.2. For every $x \in M$,

$$\operatorname{rank} L_x = 3, \qquad \operatorname{Im} L_x = F_{row(x)}.$$

Moreover,

$$\dim \operatorname{span}\{L_x : x \in M\} = 9.$$

Proof. Proposition 14.1 shows that each L_x maps onto the three-point row fiber $F_{row(x)}$, hence has rank three after linearization. Operators with different target rows have disjoint codomain support. For a fixed row, the three restrictions to the row fiber are linearly independent; in normalized row 0 they contain the three matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus each row contributes dimension 3, and the total dimension is 9. The `L3/` operator rank and spectrum audit tables record the nine-row calculation. □

Assume $\operatorname{char} k \neq 2$.

Theorem 14.3. If $x = (r, r)$, then

$$\mu_{L_x}(t) = t(t-1)(t+1)$$

and L_x is semisimple. If $x = (r, c)$ with $r \neq c$, then

$$\mu_{L_x}(t) = t^2(t-1)(t+1),$$

with zero-primary Jordan structure $J_2(0) \oplus J_1(0)^{\oplus 5}$.

Proof. Every $L_{(r,c)}$ maps $k[M]$ onto kF_r, so the nonzero spectral part is the restriction to kF_r. For $c = r$, this restriction fixes (r, r) and swaps the two off-diagonal basis vectors. For $c \neq r$, the restriction has eigenvalues $0, 1, -1$, and its one-dimensional kernel lies inside the image, producing exactly one length-two zero block. The operator-spectrum table audits the calculation for all nine translations. □

14.2 Positive composition envelope

Let $\mathrm{Mon}^+\langle L_x\rangle$ be the positive composition monoid generated by the nine left translations. For $r \in S$, let $\mathcal{W}_r$ be the component of maps whose image lies in F_r.

For fixed r, define

$$\tau_{r,c}(d) = \begin{cases} r, & d = c, \\ \overline{cd}, & d \neq c, \end{cases} \qquad \mathcal{P}_r = \langle \tau_{r,c} : c \in S\rangle.$$

In normalized row 0,

$$\mathcal{P}_0 = \{012, 021, 200, 100, 011, 022\}.$$

For $\psi \in \mathcal{P}_r$ and $c \in S$, set

$$N_{r,\psi,c}(s,d) = \begin{cases} (r, \psi(\tau_{r,c}(d))), & s = r, \\ (r, \psi(s)), & s \neq r, \end{cases}$$

and let $K_{r,t}$ be the constant map to (r,t).

Proposition 14.4 (Fixed-row composition normal form)**.** For every $r \in S$,

$$\mathcal{W}_r = \{N_{r,\psi,c} : \psi \in \mathcal{P}_r,\ c \in S\} \cup \{K_{r,t} : t \in S\}.$$

There are 14 nonconstant maps of the first type and 3 constants; hence $|\mathcal{W}_r| = 17$.

Proof. A positive composition is owned by its leftmost generator. If all later generators have the same target row r, the word has the displayed $N_{r,\psi,c}$ form. If some later generator has target row different from r, the leftmost row-r generator sees a cross-row input and the composition becomes constant. The normalized finite count $18 - 4 = 14$ for the nonconstant forms is audited in the composition envelope table. □

Theorem 14.5. The positive monoid generated by the left translations has size

$$|\,\mathrm{Mon}^+\langle L_x\rangle| = 51.$$

More precisely,

$$\mathrm{Mon}^+\langle L_x\rangle = \mathcal{W}_0 \sqcup \mathcal{W}_1 \sqcup \mathcal{W}_2, \qquad |\mathcal{W}_r| = 17.$$

Its linear envelope decomposes as

$$\mathcal{E} = W_0 \oplus W_1 \oplus W_2, \qquad \dim W_r = 9, \qquad \dim \mathcal{E} = 27,$$

and satisfies $W_r W_s \subseteq W_r$.

Proof. Proposition 14.4 gives $|\mathcal{W}_r| = 17$, and the three target-row components are disjoint. The row-owner law gives $W_r W_s \subseteq W_r$. The linear dimension claim is the row-wise basis-reduction audit recorded in the `L3/` composition-envelope and product-law tables. □

14.3 Row-fiber algebra and directed-edge representation

Assume $\mathrm{char}\, k \neq 2$. For each row r, define

$$A_{5,r} = \langle L_{(r,c)}|_{kF_r} : c \in S\rangle \subseteq \mathrm{End}(kF_r).$$

Proposition 14.6. The algebra $A_{5,r}$ has dimension 5. Its center is kI, its radical is two-dimensional and square-zero, and

$$A_{5,r}/\operatorname{Rad}(A_{5,r}) \cong k^3.$$

For $r = 0$, one basis is

$$L_{(0,0)}|_{kF_0}, \quad L_{(0,1)}|_{kF_0}, \quad L_{(0,2)}|_{kF_0}, \quad I, \quad (L_{(0,1)}|_{kF_0})^2.$$

Proof. By σ-equivariance it is enough to work in row 0. The five-dimensional multiplication table, center, radical, quotient, and operator-boundary checks are recorded in the auxiliary fiber-algebra certificate files listed in `SUPPORT_INDEX.txt`. □

Theorem 14.7. On the six directed edges,

$$\mathbb{R}[M^\times] \cong \mathbf{1} \oplus \mathrm{sign} \oplus 2\,\mathrm{std}.$$

For RP, the absorption matrix is

$$A_{abs} = (P_C + P_J)^\top$$

and has spectrum

$$\{2, 0, 0, 0, -1, -1\}.$$

Proof. The six directed edges carry the permutation representation of the directed triangle. The pure absorption transitions of RP are exactly C and J, so the absorption matrix is $(P_C + P_J)^\top$. The transition and spectrum tables audit the finite calculation. □

15 Endomorphisms, intertwiners, and matrix row-contraction

For a map $\rho : S \to S$, write $\varphi_\rho(r, c) = (\rho(r), \rho(c))$. For $t \in S$, write $e_t(r, c) = (t, t)$.

For a row r, set

$$c \star_r d = \begin{cases} r, & c = d, \\ \overline{cd}, & c \neq d. \end{cases}$$

Lemma 15.1 (row-fiber homomorphisms)**.** If $\phi : S \to S$ satisfies $\phi(r) = t$ and

$$\phi(c \star_r d) = \phi(c) \star_t \phi(d) \qquad (c, d \in S),$$

then ϕ is either the constant map to t, or a permutation of S sending r to t.

Proof. After translating to $r = t = 0$, write $\phi(1) = u$, $\phi(2) = v$. The relations $0 \star_0 1 = 2$ and $0 \star_0 2 = 1$ force

$$(u, v) = (0, 0), \quad (1, 2), \quad (2, 1),$$

which are respectively the constant solution and the two permutations fixing 0. □

Theorem 15.2. The endomorphism monoid of the selected RP magma is

$$\mathrm{End}(M, \cdot) = \{\varphi_\rho : \rho \in \mathrm{Sym}(S)\} \cup \{e_0, e_1, e_2\}.$$

Proof. The idempotents of RP are exactly the diagonal points (r, r). Thus an endomorphism sends row fibers to row fibers over a map $\rho : S \to S$, and its restriction to each row fiber is governed by Lemma 15.1. The cross-row rule forces all nonconstant row restrictions to be the same permutation ρ. If ρ fails to be injective, the cross-row equations force all affected row restrictions to collapse, and then the third row collapses as well; the map is one of the e_t. Conversely, simultaneous permutations and diagonal collapses preserve the three branches of the RP multiplication. □

For an endomorphism φ, let Hom_φ be the space of intertwiners satisfying

$$TL_x = L_{\varphi(x)}T \qquad (x \in M).$$

Let $V = k[M]$, and let

$$\varepsilon : V \to k, \qquad \varepsilon\left(\sum_{m\in M} a_m e_m\right) = \sum_{m\in M} a_m$$

be the total augmentation functional. For $r \in S$, set

$$d_r = e_{(r,r)}, \qquad o_r = \sum_{c\neq r} e_{(r,c)}, \qquad U_r = \mathrm{span}\{d_r, o_r\}.$$

Here $u\varepsilon$ denotes the rank-one operator $v \mapsto \varepsilon(v)u$.

Proposition 15.3 (constant intertwiners)**.** For each diagonal collapse $e_r(x) = (r, r)$,

$$\mathrm{Hom}_{e_r} = \{u\varepsilon : u \in U_r\} = \left\{\left(\alpha e_{(r,r)} + \beta \sum_{c\neq r} e_{(r,c)}\right)\varepsilon : \alpha, \beta \in k\right\}.$$

In particular, $\dim \mathrm{Hom}_{e_r} = 2$.

Proof. Each L_x is induced by a map $M \to M$, hence every column of its matrix has sum one and $\varepsilon L_x = \varepsilon$. If $u \in \mathrm{Fix}(L_{(r,r)})$, then

$$(u\varepsilon)L_x = u\varepsilon \qquad \text{and} \qquad L_{(r,r)}(u\varepsilon) = (L_{(r,r)}u)\varepsilon = u\varepsilon,$$

so $u\varepsilon \in \mathrm{Hom}_{e_r}$.

Conversely, if $T \in \mathrm{Hom}_{e_r}$, then

$$TL_x = L_{(r,r)}T = TL_y \qquad (x, y \in M),$$

so T kills

$$\mathcal{D} = \mathrm{span}\{(L_x - L_y)v : x, y \in M,\ v \in V\}.$$

Because $\varepsilon L_x = \varepsilon$, one has $\mathcal{D} \subseteq \ker \varepsilon$. The reverse inclusion is finite: the graph generated by relations $x \cdot z \sim y \cdot z$, with common source z, is connected. The Front-C verifier records an explicit eight-edge spanning tree, so the differences in $\mathcal{D}$ span the augmentation-zero hyperplane. Thus $\mathcal{D} = \ker \varepsilon$, and T factors uniquely as $T = u\varepsilon$.

Substitution into the intertwining equation gives $L_{(r,r)}u = u$. Finally, $L_{(r,r)}$ has image in the row fiber $F_r = \mathrm{span}\{e_{(r,c)} : c \in S\}$; on this fiber it fixes $e_{(r,r)}$ and swaps the two off-diagonal basis vectors. Therefore

$$\mathrm{Fix}(L_{(r,r)}) = \mathrm{span}\left\{e_{(r,r)}, \sum_{c\neq r} e_{(r,c)}\right\} = U_r,$$

which proves the formula. □

For automorphisms, $\mathrm{Hom}_{\varphi_\rho} = kP_\rho$ has dimension 1. The Front-C verifier audits the defining linear constraints: the collapse spaces have dimension 2, the six automorphism spaces have dimension 1, and therefore the total audit dimension is

$$6 \cdot 1 + 3 \cdot 2 = 12.$$

15.1 Matrix row-contraction bridge

Proposition 15.4. Under $k[M] \cong \mathrm{Mat}_3(k)$, the row-contraction extension of the cross-row RP rule is

$$\mu_{cross}(A, B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top.$$

On matrix units with cross rows,

$$E_{ij}J_3E_{kl}^\top = E_{ik}.$$

Proof. Since $J_3 = \mathbf{1}\mathbf{1}^\top$,

$$E_{ij}J_3E_{kl}^\top = (e_ie_j^\top)(\mathbf{1}\mathbf{1}^\top)(e_le_k^\top) = E_{ik}.$$

For $i \neq k$, this is exactly the cross-row RP rule $(i, j) \cdot (k, l) = (i, k)$. The `L3/` matrix-bridge audit tables record the basis-level bridge. □

Thus the matrix bridge expresses the same column-blindness selected by $\mathcal{H}_{acc} = 0$: cross-row multiplication factors through row sums. The same-row branch remains the Steiner/diagonal row-fiber rule.

16 Hidden-continuation operator bridge

The local hidden-continuation identity of Section 7 has an operator interpretation in the left-regular envelope. For a normalized local index q, D_C compares the two stepwise continuation signatures, while D_L compares the two collapsed left translations.

Theorem 16.1 (Hidden-continuation operator identity)**.** Let $g \in \Omega'$ be a completed normal-form magma and let $q = (b, t, a, e, f) \in S^5$. Define

$$xy, \qquad yz, \qquad z_R, \qquad ep_L, \qquad ep_R$$

from g as in Theorem 7.1. Then

$$\frac{h_q(g)}{2} = D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R).$$

For brevity write

$$D_C(q; g) = D_C(t, xy, f; b, a, z_R), \qquad D_L(q; g) = D_L(ep_L, ep_R).$$

After selection, take $g = \mathsf{RP}$ and write the normalized triple as

$$X = (0, a), \qquad Y = (b, e), \qquad Z = (t, f).$$

Then

$$XY = (0, xy), \qquad YZ = (b, z_R),$$

and

$$(XY)Z = (0, ep_L), \qquad X(YZ) = (0, ep_R).$$

With the convention $AB = A \circ B$, the continuation distance $D_C(q; \mathsf{RP})$ is the Hamming-distance contrast between the normalized signatures of

$$L_{XY}L_Z \quad \text{and} \quad L_X L_{YZ},$$

whereas $D_L(q; \mathsf{RP})$ is the Hamming-distance contrast between

$$L_{(XY)Z} \quad \text{and} \quad L_{X(YZ)}.$$

As maps, all four operators lie in $\mathcal{W}_{\mathrm{row}(X)}$; after linearization they lie in $W_{\mathrm{row}(X)}$.

Define

$$I_{tot}(g) = 6 \sum_{q \in S^5} D_C(q; g), \qquad B_{tot}(g) = 6 \sum_{q \in S^5} D_L(q; g).$$

Thus $H_{tot}(g) = I_{tot}(g) - B_{tot}(g)$. For RP,

$$I_{tot} = 8904, \qquad B_{tot} = 1884, \qquad H_{tot} = 7020, \qquad N_- = 0.$$

The finalist RC has the same pure hidden-continuation profile.

Proof. The scalar identity is exactly Theorem 7.1 divided by 2: D_C is the sum of the nine continuation disagreement indicators, D_L is the sum of the nine collapsed left-translation disagreement indicators, and h_q is twice their signed difference.

For $g = \mathsf{RP}$, the displayed normalized triple identifies the two continuation signatures with the signatures of $L_{XY}L_Z$ and $L_X L_{YZ}$, under the convention $AB = A \circ B$. Likewise the collapsed signatures are the left translations by the endpoints $(XY)Z$ and $X(YZ)$. The row-target assertion follows from the composition-envelope law $W_r W_s \subseteq W_r$. The `L3/` hidden-continuation bridge audit records the RP/RC comparison and verifies the displayed values. The global frontier status is Theorem 7.2. □

17 Binary term-function passport: $T_2(\mathsf{RP})$

A binary term function of RP is a function $M^2 \to M$ realized by a term in variables x, y. The set of all such functions is denoted $T_2(\mathsf{RP})$.

By the row-owner lemma, every binary term has an owner variable $\mathrm{ow}(f) \in \{x, y\}$. Order the inputs as $(o, n) = (\text{owner}, \text{nonowner})$. On the same-row sector, normalize the common row to 0 and define

$$s_f(i, j) = \mathrm{col}\big(f(o = (0, i), n = (0, j))\big).$$

On the cross-row sector, normalize the owner row to 0 and the nonowner row to 1; the output column is independent of the nonowner column, so

$$u_f(i) = \mathrm{col}\big(f(o = (0, i), n = (1, j))\big).$$

The opposite cross-row direction is determined by $u_f^-(i) = -u_f(-i)$.

Proposition 17.1 (Binary sector reduction)**.** Every binary term function has a normal signature

$$\Theta(f) = \big(\mathrm{ow}(f), s_f, u_f\big).$$

The same-row signature $s_f : S^2 \to S$ and cross unary signature $u_f : S \to S$ determine the term function on all row sectors.

Proof. The owner coordinate is Lemma 13.1. Same-row inputs remain in the common input row. Cross-row inputs are governed by the owner row and a unary owner-column signature; the opposite direction is forced by the RP-compatible relabelling exchanging the two nonowner directions. □

The realized same-row signatures form a 21-element set $\mathcal{S}_{21}$. The realized cross unary signatures form a 15-element set $\mathcal{U}_{15}$. The candidate normal forms are

$$\{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

Lemma 17.2 (Binary term certificate protocol)**.** The binary term theorem is supported by four finite obligations: T2-SIG exposes $\mathcal{S}_{21}$ and $\mathcal{U}_{15}$; T2-REAL realizes every triple; T2-TRANS checks closure of the two factor transition systems; T2-MULT reconstructs the full 630×630 multiplication table from the owner and factor transitions.

Theorem 17.3. The binary term-function algebra of RP is finite and satisfies

$$T_2(\mathsf{RP}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

Consequently

$$|T_2(\mathsf{RP})| = 2 \cdot 21 \cdot 15 = 630.$$

Proof. Proposition 17.1 gives injectivity of the normal signature. The realization certificate supplies one realizing term for every triple, and the finite transition tables reconstruct the product law. Hence the cardinality is $2 \cdot 21 \cdot 15$. □

This is the binary layer of the term algebra. The all-arity clone is left for the open-problem list.

18 Carrier-level matrix and Lie companion structures

This section records ordinary matrix and Lie structures on the vector-space carrier exposed after RP selection [18, 12, 17]. The bridge is carrier-level: it uses

$$k[M] \cong \mathrm{Mat}_3(k)$$

and the RP-compatible simultaneous symbol-permutation action. The finite RP multiplication and ordinary matrix multiplication are two bilinear structures on this nine-dimensional carrier.

In this section the reductive decomposition is used only over fields of characteristic different from 3, and the compact-form statements are over $\mathbb{R}$ or $\mathbb{C}$. Ordinary matrix multiplication gives

$$\mathfrak{gl}(3, k) = kI \oplus \mathfrak{sl}(3, k), \qquad \mathrm{Mat}_3(k) = kI \oplus \mathfrak{sl}(3, k)$$

as a vector-space decomposition. After complex compact form, this is the usual central $\mathfrak{u}(1)$ plus $\mathfrak{su}(3)$ picture.

Over $\mathbb{R}$, the compact three-dimensional branch

$$\mathfrak{so}(3) = \{X \in \mathfrak{sl}(3, \mathbb{R}) : X^\top = -X\}$$

is stable under simultaneous permutation conjugation. A split comparator is $\mathfrak{sl}(V_{std}) \subset \mathfrak{sl}(3, \mathbb{R})$, where $V_{std} = \{v : v_0 + v_1 + v_2 = 0\}$. Compactness selects the $\mathfrak{so}(3)$ branch among these three-dimensional $\mathrm{Sym}(S)$-stable carrier branches.

Theorem 18.1. In the real/complex compact-form setting, the carrier identification and RP-compatible $\mathrm{Sym}(S)$-action expose the following carrier-level compact companions:

$$SU(3), \qquad SU(2), \qquad U(1).$$

Here $SU(3)$ is the compact form attached to the ordinary $\mathfrak{sl}(3)$ matrix package, $SU(2)$ is the double cover of the selected compact $SO(3)$ branch, and $U(1)$ is the compact central factor attached to kI. The assertion is a carrier-level companion statement; it does not assert a canonical direct-product action $SU(3) \times SU(2) \times U(1)$ generated by the finite magma.

Proof. The matrix-unit carrier gives Mat_3, hence the usual $\mathfrak{sl}(3)$ compact form and the central circle after complex compact form. The $\mathrm{Sym}(S)$-stable compact branch $\mathfrak{so}(3)$ gives $SO(3)$, with double cover $SU(2)$. This $SO(3)$ branch is a selected compact sub-branch of the carrier matrix algebra, not an independent factor commuting with $SU(3)$. The `L3/` companion-structure bridge audit records the carrier-level compact types. □

19 Signed symplectic lift of the pure C/J geometry

This bridge uses the finite pure C/J directed-edge geometry selected in Part II. The continuous statement is a signed projective symplectic realization of that six-edge geometry and completes the carrier-level bridge part of the passport [22].

Let $V = \mathbb{R}^6$ have ordered basis

$$(e_{01}, e_{12}, e_{20}, e_{10}, e_{02}, e_{21})$$

and symplectic matrix

$$\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}.$$

Set

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\widehat{C} = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}, \qquad \widehat{J}_0 = \begin{pmatrix} 0 & R \\ R & 0 \end{pmatrix}, \qquad \widehat{J}_s = \begin{pmatrix} 0 & R \\ -R & 0 \end{pmatrix}.$$

The unsigned $\widehat{J}_0$ induces edge reversal but is anti-symplectic; the signed $\widehat{J}_s$ induces the same projective edge reversal and is symplectic.

Theorem 19.1. The finite pure $\{C, J\}$ directed-edge geometry admits the signed projective dihedral lift displayed above. The unsigned reversal is anti-symplectic, while the signed reversal is symplectic. The finite tick word

$$C, J, C, C, J$$

visits all six directed edges and has a signed symplectic realization.

Proof. Direct multiplication gives

$$\widehat{C}^T \Omega \widehat{C} = \Omega, \qquad \widehat{J}_0^T \Omega \widehat{J}_0 = -\Omega, \qquad \widehat{J}_s^T \Omega \widehat{J}_s = \Omega,$$

and

$$\widehat{C}^3 = I, \qquad \widehat{J}_s^2 = -I, \qquad \widehat{J}_s \widehat{C} \widehat{J}_s^{-1} = \widehat{C}^{-1}.$$

Hence the projective classes satisfy the dihedral relations in $\mathrm{PSp}(V,\omega)$. On coordinate lines, $\widehat{C}$ acts as (01 12 20)(10 02 21), and $\widehat{J}_s$ acts as edge reversal. Starting from 01, the tick word gives

$$01 \mapsto 12 \mapsto 21 \mapsto 10 \mapsto 02 \mapsto 20.$$

The signed-lift matrices and tick-word audit are recorded in the `L3/` signed-symplectic bridge tables. □

This completes the post-selection passport: the finite operator structure, binary term structure, matrix row-contraction, carrier-level matrix/Lie bridge, and signed symplectic lift are all read from the single selected multiplication RP and its pure directed-edge geometry.

Part IV

Certificates and open bridges

20 Finite certificate protocol

The support package is available at `https://github.com/burundai-t/finite-row-graded-magmas-over-f3`.

The repository license split is: manuscript files in `paper/` are licensed under CC BY 4.0, while all non-paper support files are licensed under MIT.

It separates three repository roles:

`L1/`, `L2/`, `L3/` frozen finite reference layers,

`mathcal_H/` the global $\mathcal{H}$-frontier bundle,

`README.md`, `SUPPORT_INDEX.txt`, `tools/`, `VERIFY_LOGS/` the repository interface.

The frozen `L1/`, `L2/`, and `L3/` layers support the manuscript-facing Landscape/Selection/Linearization finite checks. The `mathcal_H/` bundle supports the global $\mathcal{H}$-frontier theorem of Section 7, and `mathcal_H/h2/verify_h2_coverage_light.py` provides the static coverage check for the accepted H2 raw artifacts.

The reviewer-facing finite-layer check is

```
tools/run_quick_checks.sh
```

It runs the frozen L1/L2/L3 audit path:

layer	role
L1	finite landscape and associativity reference checks
L2	compact selection and pure C/J survivor checks
L3	selected-RP operator, bridge, term-algebra, and companion-structure checks

The compact table families are:

family	tables	purpose
L1 landscape atlas	`L1/tables/`	associativity distribution, endpoints, controlled $\mathcal{H}$ comparison
L2 selection	`L2/tables/`	selection chain and pure C/J survivor separation
L3 linearization passport	`L3/tables/`	operator envelope, T_2, matrix bridge, signed symplectic lift, auxiliary fiber algebra

The complete file-by-file index, including the bundle root interface and H2 light coverage support, is `SUPPORT_INDEX.txt`.

20.1 Global $\mathcal{H}$-frontier certificate handles

The global $\mathcal{H}$-frontier theorem is checked by the separate `mathcal_H/` bundle. The manuscript-level implication is

$$\text{local distance reduction} + \text{H1–H4 certificate obligations} \Longrightarrow \text{Theorem 7.2.}$$

handle	bundle location	theorem component
HF-RED, HF-CERT-IF	bundle theorem interface	local distance reduction
H1	`h1/`	unrestricted full-landscape range
H2	`h2/`	pure-frontier UNSAT certificate
H3	embedded verifier check	RP/RC pure-frontier witnesses
H4	`h4/`	signed-cancellation spike classifier

The exact repository commands for the bundle root and focused H1–H4 checks are listed in `SUPPORT_INDEX.txt`. The H2 obligation proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

UNSAT on the depth-9 live frontier:

$$18540/18540 \text{ UNSAT}, \qquad SAT = 0, \qquad UNKNOWN = 0.$$

The light static H2 coverage verifier checks the accepted raw H2 artifact coverage and integrity for the declared live frontier. This command is the artifact-coverage audit for H2:

```
cd mathcal_H/h2
PYTHONDONTWRITEBYTECODE=1 python3 verify_h2_coverage_light.py
```

20.2 Dependency summary

result family	support role
compact RP uniqueness, Theorem 12.3	hand finite proof; L2 audit tables
explanatory selection chain, Proposition 12.4	didactic chain, logically subsumed by Theorem 12.3; local audits
global associativity atlas	finite `L1/` atlas tables
global $\mathcal{H}$-frontier	separate H1–H4 certificate bundle
linearization passport	analytic/classification results with L3 audits
binary term algebra	sector reduction plus realization/transition certificates
matrix/Lie and signed symplectic structures	bridge audits

21 Open problems and bridges

The following fronts are subsequent work. The selection theorem and the operator passport above are complete at the level stated in this manuscript.

21.1 Closures

21.1.1 Functorial RP–$\mathfrak{gl}(3)$ bridge

The carrier identification

$$k[M] \cong \mathrm{Mat}_3(k)$$

supports two bilinear operations: RP multiplication and ordinary matrix multiplication. The cross-row contraction

$$\mu_{cross}(A, B) = AJ_3B^\top$$

and the carrier-level compact companions give partial alignments. The closure question is whether there is a universal or functorial construction from the RP magma to the matrix/Lie package, beyond the shared carrier.

21.1.2 Symplectic development of pure C/J

The finite pure C/J geometry has an explicit signed symplectic lift. The next problem is to develop the associated continuous symplectic geometry, including canonical (q, p)-charts, Hamiltonian paths, and coordinate-free separation of pure reversal from drifted reversal.

21.1.3 Unified selection principle

Theorem 12.3 selects RP by the conjunction of four finite selectors of distinct nature: information-theoretic, entropic, algebraic, and combinatorial-geometric. It remains open whether these selectors arise from a single natural, non-tautological information-theoretic or algebraic functional whose global minimizers on Ω' are exactly the RP representative. In such a formulation, RP would appear as a global extremum of one canonical object rather than as the intersection of four independently stated extrema.

21.2 Extensions

21.2.1 All-arity term algebra

The binary term algebra satisfies

$$|T_2(\mathsf{RP})| = 630, \qquad T_2(\mathsf{RP}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

The all-arity term algebra remains open: the main question is whether the binary factorization extends to a general clone structure or is genuinely binary [24, 27, 20].

21.2.2 Larger carriers

Natural generalizations include analogues over larger carriers $S \times S$, especially finite fields or cyclic sets, and directed-edge C/J geometries beyond the triangle.

21.2.3 Relaxing AX2

The landscape Ω' fixes the same-row Steiner axiom. The larger relaxed landscape asks whether the same-row Steiner rule can itself be selected by intrinsic criteria.

21.3 Status summary

Front	Status
Global $\mathcal{H}$-frontier certificate	closed by Theorem 7.2 and the `mathcal_H/` certificate bundle
Functorial RP–$\mathfrak{gl}(3)$ bridge	open
Unified selection principle	open
Symplectic development of pure C/J	open
All-arity term algebra	open
Larger carriers	open
Relaxing AX2	open

22 Conclusion

The RP magma is selected from the finite normal-form landscape $\Omega' \cong S^{21}$ by four internal finite criteria:

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J.$$

The surrounding landscape is mapped by the certified associativity atlas and the $\mathcal{H}$-frontier atlas, and the selected magma then exposes its linear/operator passport, binary term algebra, carrier-level compact Lie companions, and signed symplectic bridge.

Thus the theorem package has the intended three-layer form:

$$\boxed{\text{finite landscape}} \qquad \boxed{\text{intrinsic finite selection}} \qquad \boxed{\text{linearization of } \mathsf{RP}}.$$

A Information-theoretic interpretation of the selected magma

This appendix records the information-theoretic reading behind the four selectors used in Theorem 12.3 and the six basic properties of the selected magma RP. The proofs and finite classifications are in the main text; the purpose here is to keep the Landauer and Shannon motivations visible in the final mathematical form of the paper.

A.1 Two axes: economy and content

The selected magma is read along two coupled axes.

The first axis is *economy*. It records which distinctions the operation can avoid reading, storing, or drifting. The hidden-column access selector

$$\mathcal{H}_{acc} = I(C_{out}; C_1, C_2 \mid R_1, R_2)$$

implements this axis on the cross-row branch. Its zero value is exactly column-blindness: after the ordered row pair is known, the output column is constant over the hidden input columns. In the Landauer reading, a column coordinate that is read and then irreversibly discarded contributes a thermodynamic bookkeeping cost [19, 3, 4]. The condition $\mathcal{H}_{acc} = 0$ selects the rules whose cross-row output can be computed through the public row interface.

The second axis is *content*. It records which finite distinctions the operation preserves: directed public contact, bracket-sensitive histories, row-fixed self-action, and literal directed-edge continuation/reversal. The selected point is economical because the cross-row channel uses only public row data, and it is nontrivial because the right public row remains visible in the output.

The rows are therefore the public interface of the interaction, while columns are the hidden internal coordinate. The selected cross-row rule

$$(r_1, c_1) \cdot (r_2, c_2) = (r_1, r_2), \qquad r_1 \neq r_2,$$

keeps the left row as owner of the result and writes the right row as the output column.

A.2 Shannon diagonal compression

The diagonal selector is the Shannon entropy of the empirical same-point output distribution

$$H_{diag}(d) = -\sum_{s \in S} p_s \log_2 p_s.$$

Shannon entropy measures the number of bits needed to encode the diagonal output type [26, 10]. The zero-entropy condition is the finite compression rule: all same-point diagonal outputs carry one symbol. Row-anchor idempotence then chooses the normalized row base,

$$(r, c)^2 = (r, r),$$

so the compressed self-action returns to the diagonal point of the same row.

Thus $H_{diag} = 0$ is the diagonal economy selector, while row-anchor idempotence is the semantic stabilization of that compressed sector.

A.3 Six properties of RP

The following table summarizes the interpretation of the six basic properties that appear across the selection and linearization layers.

Property of RP	Axis	Physical-theoretical unfolding
$\mathcal{H}_{acc} = 0$	economy	The cross-row channel uses public rows alone. In the Landauer reading, the column-read burden is structurally avoided.
$(a, b) = (1, 2)$	content	The right public direction remains visible: $h(r_1, r_2) = r_2$. Interaction keeps a finite directional distinction while staying column-blind.
Low local Assoc_{000} among nontrivial column-blind rules	content	Bracket placement carries finite path information. The diagnostic local selector exposes the two finalist histories before the directed-edge criterion separates them.
Pure undrifted C/J	economy/content	Directed-edge absorption is literal continuation and reversal, $\mathrm{AbsTrans}_{\mathsf{RP}} = \{C, J\}$. At the linear level this finite geometry admits the signed symplectic lift of Section 19.
$H_{diag} = 0$	economy	The same-point diagonal stores one output type. This is the Shannon compression of the diagonal sector.
$(r, c)^2 = (r, r)$	content	Self-application returns to the row-fixed point. Each row has a stable diagonal anchor for self-action.

The local associativity entry is a diagnostic part of the explanatory chain, while the compact uniqueness theorem uses the four-condition core

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{row-anchor idempotence}, \qquad \text{pure } C/J.$$

After the first three conditions, pure C/J already forces the cross-row anchors $(a, b) = (1, 2)$, hence the right-row projection rule.

A.4 Variational reading

The four-condition selector can be read as a finite variational principle:

$$\boxed{\text{minimize hidden information expenditure} \quad \text{while preserving directed interaction content.}}$$

The economy side is supplied by zero hidden-column access and zero diagonal entropy. The content side is supplied by row-anchor idempotence, directed public row dependence, bracket-sensitive local histories, and pure C/J geometry.

The linear structures appearing after selection are compatible with this reading:

$$\begin{aligned}
\mathcal{H}_{acc} = 0 &\rightsquigarrow \text{column-blind row interface and row-contraction,}\\
H_{diag} = 0 &\rightsquigarrow \text{diagonal compression,}\\
(r, c)^2 = (r, r) &\rightsquigarrow \text{row-fixed self-action,}\\
(a, b) = (1, 2) &\rightsquigarrow \text{right-row projection,}\\
\text{low local } \mathrm{Assoc}_{000} &\rightsquigarrow \text{path-sensitive interaction,}\\
\text{pure } C/J &\rightsquigarrow \text{dihedral directed-edge geometry and signed symplectic lift.}
\end{aligned}$$

This semantic reading explains the post-selection passport: the operator, matrix, term-function, Lie-compatible, and signed symplectic structures are linear shadows of the same finite row/column economy and content constraints.

Note on AI assistance

This work was developed with the assistance of large language models.

The author specified the conceptual framework, constraints, and target properties of the objects under study. Within this framework, AI tools were used for computational exploration, code drafting, and structuring intermediate results.

All mathematical definitions, statements, and theorems presented in the paper were established and independently verified by the author. The finite claims are backed by exhaustive checks recorded in the accompanying support package.

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